

# Internal Dependencies

## References

- [Analyze java package metrics in a graph database](#)
- [Calculate metrics](#)
- [Neo4j Python Driver](#)

## Artifacts

List the artifacts this notebook is based on. Different sorting variations help finding artifacts by their features and support larger code bases where the list of all artifacts gets too long.

Only the top 30 entries are shown. The whole table can be found in the following CSV report:

List\_all\_Java\_artifacts

Table 1a - Top 30 artifacts with the highest package count

|    | artifactName                             | packages | types | incomingDependencies | outgoingDependencies |
|----|------------------------------------------|----------|-------|----------------------|----------------------|
| 0  | axon-messaging-5.0.2.jar                 | 57       | 570   | 7                    | 2                    |
| 1  | axon-common-5.0.2.jar                    | 15       | 156   | 10                   | 0                    |
| 2  | axon-eventsourcing-5.0.2.jar             | 8        | 104   | 3                    | 4                    |
| 3  | axon-modelling-5.0.2.jar                 | 7        | 93    | 2                    | 3                    |
| 4  | axon-spring-boot-autoconfigure-5.0.2.jar | 7        | 72    | 0                    | 7                    |
| 5  | axon-server-connector-5.0.2.jar          | 5        | 72    | 1                    | 4                    |
| 6  | axon-test-5.0.2.jar                      | 5        | 73    | 1                    | 3                    |
| 7  | axon-update-5.0.2.jar                    | 5        | 23    | 1                    | 1                    |
| 8  | axon-conversion-5.0.2.jar                | 4        | 30    | 4                    | 1                    |
| 9  | axon-metrics-micrometer-5.0.2.jar        | 2        | 13    | 0                    | 2                    |
| 10 | axon-tracing-opentelemetry-5.0.2.jar     | 1        | 5     | 0                    | 2                    |

Table 1b - Top 30 artifacts with the highest type count

|    | artifactName                             | packages | types | incomingDependencies | outgoingDependencies |
|----|------------------------------------------|----------|-------|----------------------|----------------------|
| 0  | axon-messaging-5.0.2.jar                 | 57       | 570   | 7                    | 2                    |
| 1  | axon-common-5.0.2.jar                    | 15       | 156   | 10                   | 0                    |
| 2  | axon-eventsourcing-5.0.2.jar             | 8        | 104   | 3                    | 4                    |
| 3  | axon-modelling-5.0.2.jar                 | 7        | 93    | 2                    | 3                    |
| 4  | axon-test-5.0.2.jar                      | 5        | 73    | 1                    | 3                    |
| 5  | axon-server-connector-5.0.2.jar          | 5        | 72    | 1                    | 4                    |
| 6  | axon-spring-boot-autoconfigure-5.0.2.jar | 7        | 72    | 0                    | 7                    |
| 7  | axon-conversion-5.0.2.jar                | 4        | 30    | 4                    | 1                    |
| 8  | axon-update-5.0.2.jar                    | 5        | 23    | 1                    | 1                    |
| 9  | axon-metrics-micrometer-5.0.2.jar        | 2        | 13    | 0                    | 2                    |
| 10 | axon-tracing-opentelemetry-5.0.2.jar     | 1        | 5     | 0                    | 2                    |

Table 1c - Top 30 artifacts with the highest number of incoming dependencies

The following table lists the top 30 artifacts that are used the most by other artifacts (highest count of incoming dependencies, highest in-degree).

|    | artifactName                             | packages | types | incomingDependencies | outgoingDependencies |
|----|------------------------------------------|----------|-------|----------------------|----------------------|
| 0  | axon-common-5.0.2.jar                    | 15       | 156   | 10                   | 0                    |
| 1  | axon-messaging-5.0.2.jar                 | 57       | 570   | 7                    | 2                    |
| 2  | axon-conversion-5.0.2.jar                | 4        | 30    | 4                    | 1                    |
| 3  | axon-eventsourcing-5.0.2.jar             | 8        | 104   | 3                    | 4                    |
| 4  | axon-modelling-5.0.2.jar                 | 7        | 93    | 2                    | 3                    |
| 5  | axon-server-connector-5.0.2.jar          | 5        | 72    | 1                    | 4                    |
| 6  | axon-test-5.0.2.jar                      | 5        | 73    | 1                    | 3                    |
| 7  | axon-update-5.0.2.jar                    | 5        | 23    | 1                    | 1                    |
| 8  | axon-metrics-micrometer-5.0.2.jar        | 2        | 13    | 0                    | 2                    |
| 9  | axon-spring-boot-autoconfigure-5.0.2.jar | 7        | 72    | 0                    | 7                    |
| 10 | axon-tracing-opentelemetry-5.0.2.jar     | 1        | 5     | 0                    | 2                    |

## Table 1d - Top 30 artifacts with the highest number of outgoing dependencies

The following table lists the top 30 artifacts that are depending on the highest number of other artifacts (highest count of outgoing dependencies, highest out-degree).

|    | artifactName                             | packages | types | incomingDependencies | outgoingDependencies |
|----|------------------------------------------|----------|-------|----------------------|----------------------|
| 0  | axon-spring-boot-autoconfigure-5.0.2.jar | 7        | 72    | 0                    | 7                    |
| 1  | axon-eventsourcing-5.0.2.jar             | 8        | 104   | 3                    | 4                    |
| 2  | axon-server-connector-5.0.2.jar          | 5        | 72    | 1                    | 4                    |
| 3  | axon-modelling-5.0.2.jar                 | 7        | 93    | 2                    | 3                    |
| 4  | axon-test-5.0.2.jar                      | 5        | 73    | 1                    | 3                    |
| 5  | axon-messaging-5.0.2.jar                 | 57       | 570   | 7                    | 2                    |
| 6  | axon-metrics-micrometer-5.0.2.jar        | 2        | 13    | 0                    | 2                    |
| 7  | axon-tracing-opentelemetry-5.0.2.jar     | 1        | 5     | 0                    | 2                    |
| 8  | axon-conversion-5.0.2.jar                | 4        | 30    | 4                    | 1                    |
| 9  | axon-update-5.0.2.jar                    | 5        | 23    | 1                    | 1                    |
| 10 | axon-common-5.0.2.jar                    | 15       | 156   | 10                   | 0                    |

## Table 1e - Top 30 artifacts with the lowest package count

|    | artifactName                             | packages | types | incomingDependencies | outgoingDependencies |
|----|------------------------------------------|----------|-------|----------------------|----------------------|
| 0  | axon-tracing-opentelemetry-5.0.2.jar     | 1        | 5     | 0                    | 2                    |
| 1  | axon-metrics-micrometer-5.0.2.jar        | 2        | 13    | 0                    | 2                    |
| 2  | axon-conversion-5.0.2.jar                | 4        | 30    | 4                    | 1                    |
| 3  | axon-server-connector-5.0.2.jar          | 5        | 72    | 1                    | 4                    |
| 4  | axon-test-5.0.2.jar                      | 5        | 73    | 1                    | 3                    |
| 5  | axon-update-5.0.2.jar                    | 5        | 23    | 1                    | 1                    |
| 6  | axon-modelling-5.0.2.jar                 | 7        | 93    | 2                    | 3                    |
| 7  | axon-spring-boot-autoconfigure-5.0.2.jar | 7        | 72    | 0                    | 7                    |
| 8  | axon-eventsourcing-5.0.2.jar             | 8        | 104   | 3                    | 4                    |
| 9  | axon-common-5.0.2.jar                    | 15       | 156   | 10                   | 0                    |
| 10 | axon-messaging-5.0.2.jar                 | 57       | 570   | 7                    | 2                    |

## Table 1f - Top 30 artifacts with the lowest type count

|    | artifactName                             | packages | types | incomingDependencies | outgoingDependencies |
|----|------------------------------------------|----------|-------|----------------------|----------------------|
| 0  | axon-tracing-opentelemetry-5.0.2.jar     | 1        | 5     | 0                    | 2                    |
| 1  | axon-metrics-micrometer-5.0.2.jar        | 2        | 13    | 0                    | 2                    |
| 2  | axon-update-5.0.2.jar                    | 5        | 23    | 1                    | 1                    |
| 3  | axon-conversion-5.0.2.jar                | 4        | 30    | 4                    | 1                    |
| 4  | axon-server-connector-5.0.2.jar          | 5        | 72    | 1                    | 4                    |
| 5  | axon-spring-boot-autoconfigure-5.0.2.jar | 7        | 72    | 0                    | 7                    |
| 6  | axon-test-5.0.2.jar                      | 5        | 73    | 1                    | 3                    |
| 7  | axon-modelling-5.0.2.jar                 | 7        | 93    | 2                    | 3                    |
| 8  | axon-eventsourcing-5.0.2.jar             | 8        | 104   | 3                    | 4                    |
| 9  | axon-common-5.0.2.jar                    | 15       | 156   | 10                   | 0                    |
| 10 | axon-messaging-5.0.2.jar                 | 57       | 570   | 7                    | 2                    |

## Table 1g - Top 30 artifacts with the lowest number of incoming dependencies

The following table lists the top 30 artifacts that are used the least by other artifacts (lowest count of incoming dependencies, lowest in-degree).

|    | artifactName                             | packages | types | incomingDependencies | outgoingDependencies |
|----|------------------------------------------|----------|-------|----------------------|----------------------|
| 0  | axon-metrics-micrometer-5.0.2.jar        | 2        | 13    | 0                    | 2                    |
| 1  | axon-spring-boot-autoconfigure-5.0.2.jar | 7        | 72    | 0                    | 7                    |
| 2  | axon-tracing-opentelemetry-5.0.2.jar     | 1        | 5     | 0                    | 2                    |
| 3  | axon-server-connector-5.0.2.jar          | 5        | 72    | 1                    | 4                    |
| 4  | axon-test-5.0.2.jar                      | 5        | 73    | 1                    | 3                    |
| 5  | axon-update-5.0.2.jar                    | 5        | 23    | 1                    | 1                    |
| 6  | axon-modelling-5.0.2.jar                 | 7        | 93    | 2                    | 3                    |
| 7  | axon-eventsourcing-5.0.2.jar             | 8        | 104   | 3                    | 4                    |
| 8  | axon-conversion-5.0.2.jar                | 4        | 30    | 4                    | 1                    |
| 9  | axon-messaging-5.0.2.jar                 | 57       | 570   | 7                    | 2                    |
| 10 | axon-common-5.0.2.jar                    | 15       | 156   | 10                   | 0                    |

## Table 1h - Top 30 artifacts with the lowest number of outgoing dependencies

The following table lists the top 30 artifacts that are depending on the lowest number of other artifacts (lowest count of outgoing dependencies, lowest out-degree).

|    | artifactName                             | packages | types | incomingDependencies | outgoingDependencies |
|----|------------------------------------------|----------|-------|----------------------|----------------------|
| 0  | axon-common-5.0.2.jar                    | 15       | 156   | 10                   | 0                    |
| 1  | axon-conversion-5.0.2.jar                | 4        | 30    | 4                    | 1                    |
| 2  | axon-update-5.0.2.jar                    | 5        | 23    | 1                    | 1                    |
| 3  | axon-messaging-5.0.2.jar                 | 57       | 570   | 7                    | 2                    |
| 4  | axon-metrics-micrometer-5.0.2.jar        | 2        | 13    | 0                    | 2                    |
| 5  | axon-tracing-opentelemetry-5.0.2.jar     | 1        | 5     | 0                    | 2                    |
| 6  | axon-modelling-5.0.2.jar                 | 7        | 93    | 2                    | 3                    |
| 7  | axon-test-5.0.2.jar                      | 5        | 73    | 1                    | 3                    |
| 8  | axon-eventsourcing-5.0.2.jar             | 8        | 104   | 3                    | 4                    |
| 9  | axon-server-connector-5.0.2.jar          | 5        | 72    | 1                    | 4                    |
| 10 | axon-spring-boot-autoconfigure-5.0.2.jar | 7        | 72    | 0                    | 7                    |

## Cyclic Dependencies

Cyclic dependencies occur when one package uses a class of another package and vice versa. These dependencies can lead to problems when one of these packages needs to be changed.

## Table 2a - Cyclic Dependencies Overview

Show the top 40 cyclic dependencies sorted by the most promising to resolve first. This is done by calculating the number of forward dependencies (first cycle participant to second cycle participant) in relation to backward dependencies (second cycle participant back to first cycle participant). The higher this rate (approaching 1), the easier it should be to resolve the cycle by focussing on the few backward dependencies.

Only the top 40 entries are shown. The whole table can be found in the following CSV report:

[Cyclic\\_Dependencies](#)

### Columns:

- *artifactName* identifies the artifact of the first participant of the cycle
- *packageName* identifies the package of the first participant of the cycle
- *dependentArtifactName* identifies the artifact of the second participant of the cycle
- *dependentPackageName* identifies the package of the second participant of the cycle
- *forwardToBackwardBalance* is between 0 and 1. High for many forward and few backward dependencies.
- *numberForward* contains the number of dependencies from the first participant of the cycle to the second one
- *numberBackward* contains the number of dependencies from the second participant of the cycle back to the first one
- *someForwardDependencies* lists some forward dependencies in the text format "type1 -> type2"
- *backwardDependencies* lists the backward dependencies in the format "type1 <- type2" that are recommended to get resolved

|    | artifactName                | packageName                                                           | dependentArtifactName       | dependentPackageName                                    | forwardToBackwardBalance | numberForward | numberBack |
|----|-----------------------------|-----------------------------------------------------------------------|-----------------------------|---------------------------------------------------------|--------------------------|---------------|------------|
| 0  | axon-messaging-5.0.2        | org.axonframework.messaging.core.annotation                           | axon-messaging-5.0.2        | org.axonframework.messaging.core                        | 0.959184                 | 48            |            |
| 1  | axon-messaging-5.0.2        | org.axonframework.messaging.eventhandling                             | axon-messaging-5.0.2        | org.axonframework.messaging.core                        | 0.942857                 | 34            |            |
| 2  | axon-messaging-5.0.2        | org.axonframework.messaging.eventhandling.annotation                  | axon-messaging-5.0.2        | org.axonframework.messaging.core.annotation             | 0.928571                 | 27            |            |
| 3  | axon-messaging-5.0.2        | org.axonframework.messaging.commandhandling.annotation                | axon-messaging-5.0.2        | org.axonframework.messaging.core.annotation             | 0.875000                 | 15            |            |
| 4  | axon-messaging-5.0.2        | org.axonframework.messaging.queryhandling.annotation                  | axon-messaging-5.0.2        | org.axonframework.messaging.core.annotation             | 0.857143                 | 13            |            |
| 5  | axon-modelling-5.0.2        | org.axonframework.modelling.annotation                                | axon-modelling-5.0.2        | org.axonframework.modelling                             | 0.846154                 | 12            |            |
| 6  | axon-messaging-5.0.2        | org.axonframework.messaging.core.unitofwork.transaction               | axon-messaging-5.0.2        | org.axonframework.messaging.core.unitofwork             | 0.666667                 | 5             |            |
| 7  | axon-messaging-5.0.2        | org.axonframework.messaging.eventhandling.processing.streaming.pooled | axon-messaging-5.0.2        | org.axonframework.messaging.eventhandling.configuration | 0.666667                 | 15            |            |
| 8  | axon-messaging-5.0.2        | org.axonframework.messaging.eventhandling.processing.subscribing      | axon-messaging-5.0.2        | org.axonframework.messaging.eventhandling.configuration | 0.666667                 | 15            |            |
| 9  | axon-common-5.0.2           | org.axonframework.common.configuration                                | axon-common-5.0.2           | org.axonframework.common.infra                          | 0.529412                 | 13            |            |
| 10 | axon-messaging-5.0.2        | org.axonframework.messaging.core.interception.annotation              | axon-messaging-5.0.2        | org.axonframework.messaging.core.annotation             | 0.500000                 | 15            |            |
| 11 | axon-server-connector-5.0.2 | org.axonframework.axonserver.connector.event                          | axon-server-connector-5.0.2 | org.axonframework.axonserver.connector                  | 0.428571                 | 5             |            |
| 12 | axon-server-connector-5.0.2 | org.axonframework.axonserver.connector                                | axon-server-connector-5.0.2 | org.axonframework.axonserver.connector.command          | 0.333333                 | 4             |            |
| 13 | axon-messaging-5.0.2        | org.axonframework.messaging.core.configuration                        | axon-messaging-5.0.2        | org.axonframework.messaging.eventhandling.configuration | 0.333333                 | 2             |            |
| 14 | axon-messaging-5.0.2        | org.axonframework.messaging.eventhandling.sequencing                  | axon-messaging-5.0.2        | org.axonframework.messaging.eventhandling               | 0.333333                 | 8             |            |
| 15 | axon-server-connector-5.0.2 | org.axonframework.axonserver.connector.query                          | axon-server-connector-5.0.2 | org.axonframework.axonserver.connector                  | 0.200000                 | 6             |            |
| 16 | axon-modelling-5.0.2        | org.axonframework.modelling.entity.annotation                         | axon-modelling-5.0.2        | org.axonframework.modelling.annotation                  | 0.200000                 | 3             |            |
| 17 | axon-modelling-5.0.2        | org.axonframework.modelling.entity.child                              | axon-modelling-5.0.2        | org.axonframework.modelling.entity                      | 0.142857                 | 8             |            |
| 18 | axon-messaging-5.0.2        | org.axonframework.messaging.core.unitofwork                           | axon-messaging-5.0.2        | org.axonframework.messaging.core                        | 0.111111                 | 10            |            |
| 19 | axon-server-connector-5.0.2 | org.axonframework.axonserver.connector.util                           | axon-server-connector-5.0.2 | org.axonframework.axonserver.connector                  | 0.000000                 | 2             |            |
| 20 | axon-eventsourcing-5.0.2    | org.axonframework.eventsourcing.annotation.reflection                 | axon-eventsourcing-5.0.2    | org.axonframework.eventsourcing.annotation              | 0.000000                 | 1             |            |

Table 2b - Cyclic Dependencies Break Down

Lists packages with cyclic dependencies with every dependency in a separate row sorted by the most promising dependency first.

Only the top 40 entries are shown. The whole table can be found in the following CSV report:

Cyclic\_Dependencies\_Breakdown

Columns in addition to Table 2a:

- *dependency* shows the cycle dependency in the text format "type1 -> type2" (forward) or "type2<-type1" (backward)

|    | artifactName         | packageName                                 | dependentArtifactName  | dependentPackageName             | dependency                                                                                             | forwardToBackwardBalance | n |
|----|----------------------|---------------------------------------------|------------------------|----------------------------------|--------------------------------------------------------------------------------------------------------|--------------------------|---|
| 0  | axon-messaging-5.0.2 | org.axonframework.messaging.core.annotation | axon-messaging-5.0.2   | org.axonframework.messaging.core | MessageStreamResolverUtils->FluxUtils                                                                  | 0.959184                 |   |
| 1  | axon-messaging-5.0.2 | org.axonframework.messaging.core.annotation | axon-messaging-5.0.2   | org.axonframework.messaging.core | WrappedMessageHandlingMember->Message                                                                  | 0.959184                 |   |
| 2  | axon-messaging-5.0.2 | org.axonframework.messaging.core.annotation | axon-messaging-5.0.2   | org.axonframework.messaging.core | MethodInvokingMessageHandlingMember->MessageStream\$Empty                                              | 0.959184                 |   |
| 3  | axon-messaging-5.0.2 | org.axonframework.messaging.core.annotation | axon-messaging-5.0.2   | org.axonframework.messaging.core | MessageStreamResolverUtils->MonoUtils                                                                  | 0.959184                 |   |
| 4  | axon-messaging-5.0.2 | org.axonframework.messaging.core.annotation | axon-messaging-5.0.2   | org.axonframework.messaging.core | MessageStreamResolverUtils->GenericMessage                                                             | 0.959184                 |   |
| 5  | axon-messaging-5.0.2 | org.axonframework.messaging.core.annotation | axon-messaging-5.0.2   | org.axonframework.messaging.core | MessageStreamResolverUtils->MessageStream                                                              | 0.959184                 |   |
| 6  | axon-messaging-5.0.2 | org.axonframework.messaging.core.annotation | axon-messaging-5.0.2   | org.axonframework.messaging.core | MethodInvokingMessageHandlingMember->DelayedMessageStream                                              | 0.959184                 |   |
| 7  | axon-messaging-5.0.2 | org.axonframework.messaging.core.annotation | axon-messaging-5.0.2   | org.axonframework.messaging.core | MethodInvokingMessageHandlingMember->MessageStream\$Entry                                              | 0.959184                 |   |
| 8  | axon-messaging-5.0.2 | org.axonframework.messaging.core.annotation | axon-messaging-5.0.2   | org.axonframework.messaging.core | MessageHandlingMember->MessageStream                                                                   | 0.959184                 |   |
| 9  | axon-messaging-5.0.2 | org.axonframework.messaging.core.annotation | axon-messaging-5.0.2   | org.axonframework.messaging.core | MessageStreamResolverUtils->MessageTypeResolver                                                        | 0.959184                 |   |
| 10 | axon-messaging-5.0.2 | org.axonframework.messaging.core.annotation | axondt-messaging-5.0.2 | org.axonframework.messaging.core | InterceptorChainParameterResolverFactory->Message                                                      | 0.959184                 |   |
| 11 | axon-messaging-5.0.2 | org.axonframework.messaging.core.annotation | axon-messaging-5.0.2   | org.axonframework.messaging.core | MessageStreamResolverUtils->MessageStream\$Empty                                                       | 0.959184                 |   |
| 12 | axon-messaging-5.0.2 | org.axonframework.messaging.core.annotation | axon-messaging-5.0.2   | org.axonframework.messaging.core | InterceptorChainParameterResolverFactory->MessageStream                                                | 0.959184                 |   |
| 13 | axon-messaging-5.0.2 | org.axonframework.messaging.core.annotation | axon-messaging-5.0.2   | org.axonframework.messaging.core | InterceptorChainParameterResolverFactory->MessageHandlerInterceptorChain                               | 0.959184                 |   |
| 14 | axon-messaging-5.0.2 | org.axonframework.messaging.core.annotation | axon-messaging-5.0.2   | org.axonframework.messaging.core | AggregateTypeParameterResolverFactory\$AggregateTypeParameterResolver->LegacyResources                 | 0.959184                 |   |
| 15 | axon-messaging-5.0.2 | org.axonframework.messaging.core.annotation | axon-messaging-5.0.2   | org.axonframework.messaging.core | MethodInvokingMessageHandlingMember->MessageStream                                                     | 0.959184                 |   |
| 16 | axon-messaging-5.0.2 | org.axonframework.messaging.core.annotation | axon-messaging-5.0.2   | org.axonframework.messaging.core | MethodInvokingMessageHandlingMember->Message                                                           | 0.959184                 |   |
| 17 | axon-messaging-5.0.2 | org.axonframework.messaging.core.annotation | axon-messaging-5.0.2   | org.axonframework.messaging.core | ChainedMessageHandlerInterceptorMember->MessageStream                                                  | 0.959184                 |   |
| 18 | axon-messaging-5.0.2 | org.axonframework.messaging.core.annotation | axon-messaging-5.0.2   | org.axonframework.messaging.core | AggregateTypeParameterResolverFactory<br><i>AggregateTypeParameterResolver--&gt;ContextResourceKey</i> | 0.959184                 |   |
| 19 | axon-messaging-5.0.2 | org.axonframework.messaging.core.annotation | axon-messaging-5.0.2   | org.axonframework.messaging.core | MethodInvokingMessageHandlingMember->MessageStream\$Single                                             | 0.959184                 |   |
| 20 | axon-messaging-5.0.2 | org.axonframework.messaging.core.annotation | axon-messaging-5.0.2   | org.axonframework.messaging.core | HandlerAttributes<-SimpleHandlerAttributes                                                             | 0.959184                 |   |
| 21 | axon-messaging-5.0.2 | org.axonframework.messaging.core.annotation | axon-messaging-5.0.2   | org.axonframework.messaging.core | MessageHandlingMember->Message                                                                         | 0.959184                 |   |
| 22 | axon-messaging-5.0.2 | org.axonframework.messaging.core.annotation | axon-messaging-5.0.2   | org.axonframework.messaging.core | AnnotationMessageTypeResolver->MessageTypeResolver                                                     | 0.959184                 |   |
| 23 | axon-messaging-5.0.2 | org.axonframework.messaging.core.annotation | axon-messaging-5.0.2   | org.axonframework.messaging.core | DefaultParameterResolverFactory->Message                                                               | 0.959184                 |   |
| 24 | axon-messaging-5.0.2 | org.axonframework.messaging.core.annotation | axon-messaging-5.0.2   | org.axonframework.messaging.core | AnnotationMessageTypeResolver->MessageType                                                             | 0.959184                 |   |
| 25 | axon-messaging-5.0.2 | org.axonframework.messaging.core.annotation | axon-messaging-5.0.2   | org.axonframework.messaging.core | MessageStreamResolverUtils->MessageStream\$Single                                                      | 0.959184                 |   |
| 26 | axon-messaging-5.0.2 | org.axonframework.messaging.core.annotation | axon-messaging-5.0.2   | org.axonframework.messaging.core | DefaultParameterResolverFactory->Metadata                                                              | 0.959184                 |   |
| 27 | axon-messaging-5.0.2 | org.axonframework.messaging.core.annotation | axon-messaging-5.0.2   | org.axonframework.messaging.core | AnnotationMessageTypeResolver->ClassBasedMessageTypeResolver                                           | 0.959184                 |   |
| 28 | axon-messaging-5.0.2 | org.axonframework.messaging.core.annotation | axon-messaging-5.0.2   | org.axonframework.messaging.core | PayloadParameterResolver->Message                                                                      | 0.959184                 |   |
| 29 | axon-messaging-5.0.2 | org.axonframework.messaging.core.annotation | axon-messaging-5.0.2   | org.axonframework.messaging.core | InterceptorChainParameterResolverFactory->Context\$ResourceKey                                         | 0.959184                 |   |
| 30 | axon-messaging-5.0.2 | org.axonframework.messaging.core.annotation | axon-messaging-5.0.2   | org.axonframework.messaging.core | DefaultParameterResolverFactory\$MetadataParameterResolver->Message                                    | 0.959184                 |   |
| 31 | axon-messaging-5.0.2 | org.axonframework.messaging.core.annotation | axon-messaging-5.0.2   | org.axonframework.messaging.core | AnnotatedHandlerAttributes->SimpleHandlerAttributes                                                    | 0.959184                 |   |
| 32 | axon-messaging-5.0.2 | org.axonframework.messaging.core.annotation | axon-messaging-5.0.2   | org.axonframework.messaging.core | MessageIdentifierParameterResolverFactory\$MessageIdentifierParameterResolver->Message                 | 0.959184                 |   |
| 33 | axon-messaging-5.0.2 | org.axonframework.messaging.core.annotation | axon-messaging-5.0.2   | org.axonframework.messaging.core | MultiHandlerDefinition->MessageStream                                                                  | 0.959184                 |   |
| 34 | axon-messaging-5.0.2 | org.axonframework.messaging.core.annotation | axon-messaging-5.0.2   | org.axonframework.messaging.core | DefaultParameterResolverFactory\$MetadataParameterResolver->Metadata                                   | 0.959184                 |   |
| 35 | axon-messaging-5.0.2 | org.axonframework.messaging.core.annotation | axon-messaging-5.0.2   | org.axonframework.messaging.core | ChainedMessageHandlerInterceptorMember->Message                                                        | 0.959184                 |   |
| 36 | axon-messaging-5.0.2 | org.axonframework.messaging.core.annotation | axon-messaging-5.0.2   | org.axonframework.messaging.core | AnnotatedMessageHandlingMemberDefinition->MessageStream                                                | 0.959184                 |   |
| 37 | axon-messaging-5.0.2 | org.axonframework.messaging.core.annotation | axon-messaging-5.0.2   | org.axonframework.messaging.core | AnnotatedMessageHandlingMemberDefinition->Message                                                      | 0.959184                 |   |
| 38 | axon-messaging-5.0.2 | org.axonframework.messaging.core.annotation | axon-messaging-5.0.2   | org.axonframework.messaging.core | MessageHandler->Message                                                                                | 0.959184                 |   |
| 39 | axon-messaging-5.0.2 | org.axonframework.messaging.core.annotation | axon-messaging-5.0.2   | org.axonframework.messaging.core | WrappedMessageHandlingMember->MessageStream                                                            | 0.959184                 |   |

Table 2c - Cyclic Dependencies Break Down - Backward Dependencies Only

Lists packages with cyclic dependencies with every dependency in a separate row sorted by the most promising dependency first. This table only contains the backward dependencies from the second participant of the cycle back to the first one that are the most promising to resolve.

Only the top 40 entries are shown. The whole table can be found in the following CSV report:

Cyclic\_Dependencies\_Breakdown\_BackwardOnly

|    | artifactName                | packageName                                                           | dependentArtifactName       | dependentPackageName                                    | dependency forName                                                           |
|----|-----------------------------|-----------------------------------------------------------------------|-----------------------------|---------------------------------------------------------|------------------------------------------------------------------------------|
| 0  | axon-messaging-5.0.2        | org.axonframework.messaging.core.annotation                           | axon-messaging-5.0.2        | org.axonframework.messaging.core                        | HandlerAttributes<-SimpleHandlerAttributes                                   |
| 1  | axon-messaging-5.0.2        | org.axonframework.messaging.eventhandling                             | axon-messaging-5.0.2        | org.axonframework.messaging.core                        | EventMessage<-SubscribableEventSource                                        |
| 2  | axon-messaging-5.0.2        | org.axonframework.messaging.eventhandling.annotation                  | axon-messaging-5.0.2        | org.axonframework.messaging.core.annotation             | EventHandler<-HandlerTypeResolver                                            |
| 3  | axon-messaging-5.0.2        | org.axonframework.messaging.commandhandling.annotation                | axon-messaging-5.0.2        | org.axonframework.messaging.core.annotation             | CommandHandler<-HandlerTypeResolver                                          |
| 4  | axon-messaging-5.0.2        | org.axonframework.messaging.queryhandling.annotation                  | axon-messaging-5.0.2        | org.axonframework.messaging.core.annotation             | QueryHandler<-HandlerTypeResolver                                            |
| 5  | axon-modelling-5.0.2        | org.axonframework.modelling.annotation                                | axon-modelling-5.0.2        | org.axonframework.modelling                             | TargetEntityIdMemberMismatchException<-PropertyBasedEntityIdResolver         |
| 6  | axon-messaging-5.0.2        | org.axonframework.messaging.core.unitofwork.transaction               | axon-messaging-5.0.2        | org.axonframework.messaging.core.unitofwork             | TransactionManager<-TransactionalUnitOfWorkFactory                           |
| 7  | axon-messaging-5.0.2        | org.axonframework.messaging.eventhandling.processing.streaming.pooled | axon-messaging-5.0.2        | org.axonframework.messaging.eventhandling.configuration | PooledStreamingEventProcessorConfiguration<-EventProcessorModule             |
| 8  | axon-messaging-5.0.2        | org.axonframework.messaging.eventhandling.processing.streaming.pooled | axon-messaging-5.0.2        | org.axonframework.messaging.eventhandling.configuration | PooledStreamingEventProcessorsConfigurer<-EventProcessingConfigurer          |
| 9  | axon-messaging-5.0.2        | org.axonframework.messaging.eventhandling.processing.streaming.pooled | axon-messaging-5.0.2        | org.axonframework.messaging.eventhandling.configuration | PooledStreamingEventProcessorModule<-EventProcessorModule                    |
| 10 | axon-messaging-5.0.2        | org.axonframework.messaging.eventhandling.processing.subscribing      | axon-messaging-5.0.2        | org.axonframework.messaging.eventhandling.configuration | SubscribingEventProcessorConfiguration<-EventProcessorModule                 |
| 11 | axon-messaging-5.0.2        | org.axonframework.messaging.eventhandling.processing.subscribing      | axon-messaging-5.0.2        | org.axonframework.messaging.eventhandling.configuration | SubscribingEventProcessorModule<-EventProcessorModule                        |
| 12 | axon-messaging-5.0.2        | org.axonframework.messaging.eventhandling.processing.subscribing      | axon-messaging-5.0.2        | org.axonframework.messaging.eventhandling.configuration | SubscribingEventProcessorsConfigurer<-EventProcessingConfigurer              |
| 13 | axon-common-5.0.2           | org.axonframework.common.configuration                                | axon-common-5.0.2           | org.axonframework.common.infra                          | Component\$Identifier<-JacksonComponentDescriptor                            |
| 14 | axon-common-5.0.2           | org.axonframework.common.configuration                                | axon-common-5.0.2           | org.axonframework.common.infra                          | Component<-JacksonComponentDescriptor                                        |
| 15 | axon-common-5.0.2           | org.axonframework.common.configuration                                | axon-common-5.0.2           | org.axonframework.common.infra                          | Component\$Identifier<-FilesystemStyleComponentDescriptor                    |
| 16 | axon-common-5.0.2           | org.axonframework.common.configuration                                | axon-common-5.0.2           | org.axonframework.common.infra                          | Component<-FilesystemStyleComponentDescriptor                                |
| 17 | axon-messaging-5.0.2        | org.axonframework.messaging.core.interception.annotation              | axon-messaging-5.0.2        | org.axonframework.messaging.core.annotation             | NoMoreInterceptors<-AnnotatedHandlerInspector                                |
| 18 | axon-messaging-5.0.2        | org.axonframework.messaging.core.interception.annotation              | axon-messaging-5.0.2        | org.axonframework.messaging.core.annotation             | MessageInterceptingMember<-AnnotatedHandlerInspector                         |
| 19 | axon-messaging-5.0.2        | org.axonframework.messaging.core.interception.annotation              | axon-messaging-5.0.2        | org.axonframework.messaging.core.annotation             | NoMoreInterceptors<-ChainedMessageHandlerInterceptorMember                   |
| 20 | axon-messaging-5.0.2        | org.axonframework.messaging.core.interception.annotation              | axon-messaging-5.0.2        | org.axonframework.messaging.core.annotation             | MessageHandlerInterceptorMemberChain<-ChainedMessageHandlerInterceptorMember |
| 21 | axon-messaging-5.0.2        | org.axonframework.messaging.core.interception.annotation              | axon-messaging-5.0.2        | org.axonframework.messaging.core.annotation             | MessageHandlerInterceptorMemberChain<-AnnotatedHandlerInspector              |
| 22 | axon-server-connector-5.0.2 | org.axonframework.axonserver.connector.event                          | axon-server-connector-5.0.2 | org.axonframework.axonserver.connector                  | EventProcessorControlService<-AxonServerConfigurationEnhancer                |
| 23 | axon-server-connector-5.0.2 | org.axonframework.axonserver.connector.event                          | axon-server-connector-5.0.2 | org.axonframework.axonserver.connector                  | AxonServerEventStorageEngineFactory<-AxonServerConfigurationEnhancer         |
| 24 | axon-server-connector-5.0.2 | org.axonframework.axonserver.connector                                | axon-server-connector-5.0.2 | org.axonframework.axonserver.connector.command          | AxonServerConfiguration<-AxonServerCommandBusConnector                       |
| 25 | axon-server-connector-5.0.2 | org.axonframework.axonserver.connector                                | axon-server-connector-5.0.2 | org.axonframework.axonserver.connector.command          | MetadataConverter<-CommandConverter                                          |
| 26 | axon-messaging-5.0.2        | org.axonframework.messaging.core.configuration                        | axon-messaging-5.0.2        | org.axonframework.messaging.eventhandling.configuration | MessagingConfigurer<-EventProcessingConfigurer                               |
| 27 | axon-messaging-5.0.2        | org.axonframework.messaging.eventhandling.sequencing                  | axon-messaging-5.0.2        | org.axonframework.messaging.eventhandling               | SequencingPolicy<-SimpleEventHandlingComponent                               |
| 28 | axon-messaging-5.0.2        | org.axonframework.messaging.eventhandling.sequencing                  | axon-messaging-5.0.2        | org.axonframework.messaging.eventhandling               | HierarchicalSequencingPolicy<-SimpleEventHandlingComponent                   |
| 29 | axon-messaging-5.0.2        | org.axonframework.messaging.eventhandling.sequencing                  | axon-messaging-5.0.2        | org.axonframework.messaging.eventhandling               | SequentialPolicy<-SimpleEventHandlingComponent                               |
| 30 | axon-messaging-5.0.2        | org.axonframework.messaging.eventhandling.sequencing                  | axon-messaging-5.0.2        | org.axonframework.messaging.eventhandling               | SequentialPerAggregatePolicy<-SimpleEventHandlingComponent                   |
| 31 | axon-server-connector-5.0.2 | org.axonframework.axonserver.connector.query                          | axon-server-connector-5.0.2 | org.axonframework.axonserver.connector                  | AxonServerNonTransientRemoteQueryHandlingException<-ErrorCode                |
| 32 | axon-server-connector-5.0.2 | org.axonframework.axonserver.connector.query                          | axon-server-connector-5.0.2 | org.axonframework.axonserver.connector                  | AxonServerRemoteQueryHandlingException<-ErrorCode                            |
| 33 | axon-server-connector-5.0.2 | org.axonframework.axonserver.connector.query                          | axon-server-connector-5.0.2 | org.axonframework.axonserver.connector                  | AxonServerQueryBusConnector<-AxonServerConfigurationEnhancer                 |
| 34 | axon-server-connector-5.0.2 | org.axonframework.axonserver.connector.query                          | axon-server-connector-5.0.2 | org.axonframework.axonserver.connector                  | AxonServerQueryDispatchException<-ErrorCode                                  |
| 35 | axon-modelling-5.0.2        | org.axonframework.modelling.entity.annotation                         | axon-modelling-5.0.2        | org.axonframework.modelling.annotation                  | AnnotatedEntityMetamodel<-AnnotationBasedEntityIdResolverDefinition          |
| 36 | axon-modelling-5.0.2        | org.axonframework.modelling.entity.annotation                         | axon-modelling-5.0.2        | org.axonframework.modelling.annotation                  | AnnotatedEntityMetamodel<-EntityIdResolverDefinition                         |
| 37 | axon-modelling-5.0.2        | org.axonframework.modelling.entity.child                              | axon-modelling-5.0.2        | org.axonframework.modelling.entity                      | EntityChildMetamodel<-PolymorphicEntityMetamodelBuilder                      |
| 38 | axon-modelling-5.0.2        | org.axonframework.modelling.entity.child                              | axon-modelling-5.0.2        | org.axonframework.modelling.entity                      | EntityChildMetamodel<-EntityMetamodelBuilder                                 |
| 39 | axon-modelling-5.0.2        | org.axonframework.modelling.entity.child                              | axon-modelling-5.0.2        | org.axonframework.modelling.entity                      | EntityChildMetamodel<-ConcreteEntityMetamodel\$Builder                       |



# Interface Segregation Candidates

Well known from [Design Principles and Design Patterns by Robert C. Martin](#), the *Interface Segregation Principle* suggests that software components should have narrow, focused interfaces rather than large, general-purpose ones. The goal is to minimize the dependencies between components and increase modularity, flexibility, and maintainability.

Smaller, focused and purpose-driven interfaces

- make it easier to modify individual components without affecting the rest of the system.
- make it clearer which client is affected by which change.
- don't force their clients to depend on methods they don't need.
- reduce the scope of changes since a change to one component doesn't affect others.
- lead to a more loosely coupled architecture that is easier to understand and maintain.

Reference: [Analyze java package metrics in a graph database](#)

## How to apply the results

If just one method of a type is used, especially in many places, then the result of this method can be used to call e.g. a method or construct an object instead of using the whole object and then just calling that single method.

If there are a couple of methods that are used for a distinct purpose, those could be factored out into a separate interface. The original type can extended/implement the new interface so that there are no breaking changes. Then all the callers, that use only this group of methods, can be changed to the new interface.

## Table 4 - Top 40 most used combinations of methods

The following table shows the top 40 most used combinations of methods of larger types that might benefit from applying the *Interface Segregation Principle*. The whole table can be found in the CSV report `Candidates_for_Interface_Segregation`.

|    | fullDependentTypeName                                                                                  | declaredMethods | calledMethodNames                                                                           | calledMethods | callerTypes |
|----|--------------------------------------------------------------------------------------------------------|-----------------|---------------------------------------------------------------------------------------------|---------------|-------------|
| 0  | org.axonframework.messaging.core.unitofwork.ProcessingContext                                          | 32              | [computeResourceIfAbsent]                                                                   | 1             | 7           |
| 1  | org.axonframework.common.configuration.ComponentDefinitions\$ComponentCreator                          | 17              | [createComponent]                                                                           | 1             | 5           |
| 2  | org.axonframework.messaging.core.MessageStream\$Entry                                                  | 7               | [message]                                                                                   | 1             | 5           |
| 3  | org.axonframework.messaging.commandhandling.CommandBus                                                 | 6               | [dispatch]                                                                                  | 1             | 5           |
| 4  | org.axonframework.messaging.core.unitofwork.ProcessingContext                                          | 32              | [withResource]                                                                              | 1             | 4           |
| 5  | org.axonframework.messaging.eventhandling.EventMessage                                                 | 19              | [timestamp]                                                                                 | 1             | 4           |
| 6  | org.axonframework.messaging.eventhandling.EventMessage                                                 | 19              | [timestamp, identifier]                                                                     | 2             | 4           |
| 7  | org.axonframework.messaging.core.MessageStream\$Empty                                                  | 44              | [cast]                                                                                      | 1             | 3           |
| 8  | org.axonframework.messaging.core.DelayedMessageStream                                                  | 42              | [create]                                                                                    | 1             | 3           |
| 9  | org.axonframework.messaging.eventhandling.EventMessage                                                 | 19              | [identifier]                                                                                | 1             | 3           |
| 10 | org.axonframework.messaging.core.annotation.WrappedMessageHandlingMember                               | 13              | [handleSync]                                                                                | 1             | 3           |
| 11 | org.axonframework.common.configuration.AbstractComponent                                               | 24              | [describeTo]                                                                                | 1             | 2           |
| 12 | org.axonframework.messaging.core.unitofwork.UnitOfWork                                                 | 24              | [execute]                                                                                   | 1             | 2           |
| 13 | org.axonframework.test.fixture.AxonTestThenMessage                                                     | 22              | [exception, exceptionSatisfies]                                                             | 3             | 2           |
| 14 | org.axonframework.messaging.commandhandling.CommandMessage                                             | 20              | [routingKey, priority]                                                                      | 2             | 2           |
| 15 | org.axonframework.messaging.commandhandling.CommandMessage                                             | 19              | [withConvertedPayload]                                                                      | 1             | 2           |
| 16 | org.axonframework.messaging.commandhandling.CommandMessage                                             | 19              | [andMetadata]                                                                               | 1             | 2           |
| 17 | org.axonframework.messaging.eventhandling.EventMessage                                                 | 19              | [andMetadata]                                                                               | 1             | 2           |
| 18 | org.axonframework.messaging.eventhandling.EventMessage                                                 | 19              | [withConvertedPayload]                                                                      | 1             | 2           |
| 19 | org.axonframework.messaging.eventhandling.annotation.EventHandlingMember                               | 13              | [eventName]                                                                                 | 1             | 2           |
| 20 | org.axonframework.messaging.eventstreaming.EventCriterion                                              | 12              | [tags]                                                                                      | 1             | 2           |
| 21 | org.axonframework.messaging.eventstreaming.OrEventCriteria                                             | 12              | [or]                                                                                        | 1             | 2           |
| 22 | org.axonframework.modelling.entity.PolymorphicEntityMetamodel                                          | 11              | [forSuperType]                                                                              | 1             | 2           |
| 23 | org.axonframework.modelling.entity.annotation.AnnotatedEntityMetamodel                                 | 11              | [entityType]                                                                                | 1             | 2           |
| 24 | org.axonframework.eventsourcing.eventstore.EventStore                                                  | 9               | [transaction]                                                                               | 1             | 2           |
| 25 | org.axonframework.messaging.eventhandling.processing.streaming.token.WrappedToken                      | 9               | [unwrapLowerBound]                                                                          | 1             | 2           |
| 26 | org.axonframework.messaging.eventhandling.processing.streaming.token.store.ConfigToken                 | 9               | [get]                                                                                       | 1             | 2           |
| 27 | org.axonframework.messaging.core.MessageStream\$Entry                                                  | 7               | [map]                                                                                       | 1             | 2           |
| 28 | org.axonframework.messaging.core.unitofwork.UnitOfWork\$UnitOfWorkProcessingContext                    | 45              | [whenComplete, isCompleted, commit, isError, on, isStarted, isCommitted, onError]           | 8             | 1           |
| 29 | org.axonframework.messaging.core.EmptyMessageStream                                                    | 44              | [instance]                                                                                  | 1             | 1           |
| 30 | org.axonframework.eventsourcing.eventstore.inmemory.InMemoryEventStorageEngine\$MapBackedMessageStream | 43              | [hasNextAvailable, isCompleted]                                                             | 2             | 1           |
| 31 | org.axonframework.eventsourcing.eventstore.inmemory.InMemoryEventStorageEngine\$MapBackedMessageStream | 42              | [callback]                                                                                  | 1             | 1           |
| 32 | org.axonframework.messaging.core.DelayedMessageStream                                                  | 42              | [createSingle]                                                                              | 1             | 1           |
| 33 | org.axonframework.messaging.core.MessageStream\$Single                                                 | 42              | [first]                                                                                     | 1             | 1           |
| 34 | org.axonframework.messaging.core.MessageStream\$Single                                                 | 42              | [asCompletableFuture]                                                                       | 1             | 1           |
| 35 | org.axonframework.messaging.core.MessageStream\$Single                                                 | 42              | [cast]                                                                                      | 1             | 1           |
| 36 | org.axonframework.messaging.core.unitofwork.ProcessingContext                                          | 37              | [computeResourceIfAbsent, updateResource, removeResource, putResourceIfAbsent, putResource] | 6             | 1           |
| 37 | org.axonframework.messaging.core.unitofwork.ProcessingContext                                          | 36              | [computeResourceIfAbsent, updateResource]                                                   | 2             | 1           |
| 38 | org.axonframework.messaging.core.unitofwork.ProcessingContext                                          | 35              | [computeResourceIfAbsent]                                                                   | 1             | 1           |
| 39 | org.axonframework.messaging.core.unitofwork.ProcessingContext                                          | 34              | [computeResourceIfAbsent]                                                                   | 1             | 1           |

## Package Usage

Table 5 - Types that are used by multiple packages

This table shows the top 40 packages that are used by the highest number of different packages. The whole table can be found in the CSV report `List_types_that_are_used_by_many_different_packages` .

|    | fullQualifiedDependentTypeName                                | dependentTypeName    | dependentTypeLabels                                                                                                                                                                                                                                                                                         | numberOfUsingPackages |
|----|---------------------------------------------------------------|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| 0  | org.axonframework.messaging.core.unitofwork.ProcessingContext | ProcessingContext    | [Type, File, Java, ByteCode, Interface, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityBetweenness, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mar... | 58                    |
| 1  | org.axonframework.common.annotation.Internal                  | Internal             | [Type, File, Java, ByteCode, Annotation, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent... | 51                    |
| 2  | org.axonframework.messaging.core.Message                      | Message              | [Type, File, Java, ByteCode, Interface, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityBetweenness, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mar... | 46                    |
| 3  | org.axonframework.common.infra.ComponentDescriptor            | ComponentDescriptor  | [Type, File, Java, ByteCode, Interface, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0... | 44                    |
| 4  | org.axonframework.messaging.eventhandling.EventMessage        | EventMessage         | [Type, File, Java, ByteCode, Interface, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityBetweenness, Mark4TopCentralityHarmonic, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedCompenen... | 35                    |
| 5  | org.axonframework.messaging.core.MessageStream                | MessageStream        | [Type, File, Java, ByteCode, Interface, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityBetweenness, Mark4TopCentralityHarmonic, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWea...                     | 35                    |
| 6  | org.axonframework.messaging.core.MessageType                  | MessageType          | [Type, File, Java, ByteCode, Record, Mark4TopCentralityBetweenness, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation6, M... | 30                    |
| 7  | org.axonframework.messaging.core.QualifiedName                | QualifiedName        | [Type, File, Java, ByteCode, Record, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityBetweenness, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4T... | 30                    |
| 8  | org.axonframework.common.configuration.Configuration          | Configuration        | [Type, File, Java, ByteCode, Interface, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation8, Mark4TypeLouvainCommunity6...  | 28                    |
| 9  | org.axonframework.common.FutureUtils                          | FutureUtils          | [Type, File, Java, ByteCode, Class, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation5, Mark4TypeLouvainCommunity5, Mark4... | 28                    |
| 10 | org.axonframework.messaging.core.Context\$ResourceKey         | Context\$ResourceKey | [Type, File, Java, ByteCode, GenericDeclaration, Class, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyCon... | 24                    |
| 11 | org.axonframework.common.infra.DescribableComponent           | DescribableComponent | [Type, File, Java, ByteCode, Interface, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0... | 24                    |
| 12 | org.axonframework.messaging.core.MessageStream\$Empty         | MessageStream\$Empty | [Type, File, Java, ByteCode, Interface, GenericDeclaration, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation11, Mark4TypeLouvainCommunity1, Mark4TypeLeidenCommunity8, Mark4TypeK... | 24                    |
| 13 | org.axonframework.common.Assert                               | Assert               | [Type, File, Java, ByteCode, Class, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation2, Mark4TypeLouvainCommunity1, Mark4TypeLeidenCommunity9, Mark4TypeKCoreDecomposit...  | 21                    |
| 14 | org.axonframework.common.BuilderUtils                         | BuilderUtils         | [Type, File, Java, ByteCode, Class, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation24, Mark4TypeLouvainCommunity13, Mark4TypeLeidenCommunity13, Mark4TypeKCoreDecomposition8, Mark4TypeMaximumKCUT71, Mark4TypeHDBSCAN65, Mark4TopAnomalyHub]                                                 | 21                    |
| 15 | org.axonframework.common.configuration.ComponentRegistry      | ComponentRegistry    | [Type, File, Java, ByteCode, Interface, Mark4TopCentralityBetweenness, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation16, Mark4TypeLouvainCommunity6, Mark4TypeLeidenCommunity6, Mark4TypeKCoreDecomposition10, Mark4TypeMaximumKCUT27, Mark4TypeHDBSCAN147]                                  | 21                    |
| 16 | org.axonframework.messaging.commandhandling.CommandMessage    | CommandMessage       | [Type, File, Java, ByteCode, Interface, Mark4TopCentralityArticleRank, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation2, Mark4TypeLouvainCommunity3, Mark4TypeLeidenCommunity3, ... | 20                    |
| 17 | org.axonframework.messaging.core.MessageStream\$Entry         | MessageStream\$Entry | [Type, File, Java, ByteCode, Interface, GenericDeclaration, Mark4TopCentralityArticleRank, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority,                                                                                                                                                          | 19                    |

|    | fullQualifiedDependentTypeName                                                            | dependentTypeName                                  | dependentTypeLabels                                                                                                                                                                                                                                                                                         | numberOfUsingPackages |
|----|-------------------------------------------------------------------------------------------|----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
|    |                                                                                           |                                                    | Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation11, Mark4TypeLouvainCommunity1, Mark4T...                                                                                                                                                    |                       |
| 18 | org.axonframework.messaging.core.annotation.ParameterResolverFactory                      | ParameterResolverFactory                           | [Type, File, Java, ByteCode, Interface, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation23, Mark4TypeLouvainCommunity12, Mark4TypeLeidenCommunity6, Mark4TypeKCoreDecomposition9, Mark4TypeMaximumKCut10, Mark4TypeHDBSCAN70]                                                                  | 19                    |
| 19 | org.axonframework.messaging.core.MessageStream\$Single                                    | MessageStream\$Single                              | [Type, File, Java, ByteCode, Interface, Mark4TopCentralityArticleRank, Mark4TopCentralityBetweenness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation11, Mar...                     | 18                    |
| 20 | org.axonframework.messaging.core.MessageTypeResolver                                      | MessageTypeResolver                                | [Type, File, Java, ByteCode, Interface, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation6, Mark4TypeLouvainCommunity2, Mark4TypeLeidenCommunity2, Mark4TypeKCoreDecomposition10, ... | 18                    |
| 21 | org.axonframework.common.configuration.ConfigurationEnhancer                              | ConfigurationEnhancer                              | [Type, File, Java, ByteCode, Interface, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation17, Mark4TypeLouvainCommunity6, Mark4TypeLeidenCommunity6, Mark4TypeKCoreDecomposition8, Mark4TypeMaximumKCut76, Mark4TypeHDBSCAN-1]                                                                   | 17                    |
| 22 | org.axonframework.messaging.core.conversion.MessageConverter                              | MessageConverter                                   | [Type, File, Java, ByteCode, Interface, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation8, Mark4TypeLouvainCommunity2, Mark4TypeLeidenCommunity3, Mark4TypeKCoreDecomposition9, Mark4TypeMaximumKCut20, Mark4TypeHDBSCAN136]                                                                   | 16                    |
| 23 | org.axonframework.messaging.core.Metadata                                                 | Metadata                                           | [Type, File, Java, ByteCode, Class, Mark4TopCentralityPageRank, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation0, Mark4TypeLouvainCommunity2, Mark4TypeLeidenCommunity2, Mark4TypeKCoreDecomposition9, Mark4TypeMaximumKCut11, Ma... | 16                    |
| 24 | org.axonframework.common.ObjectUtils                                                      | ObjectUtils                                        | [Type, File, Java, ByteCode, Class, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation6, Mark4TypeLouvainCommunity2, Mark4TypeLeidenCommunity2, Mark4TypeKCoreDecomposition9, Mark4TypeMaximumKCut80, Mark4TypeHDBSCAN43]                                                                        | 16                    |
| 25 | org.axonframework.messaging.eventhandling.processing.streaming.token.TrackingToken        | TrackingToken                                      | [Type, File, Java, ByteCode, Interface, Mark4TopCentralityArticleRank, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation7, Mark4TypeLouvainCommunity5, Mark4TypeLeidenCommunity5, ... | 15                    |
| 26 | org.axonframework.common.AxonConfigurationException                                       | AxonConfigurationException                         | [Type, File, Java, ByteCode, Class, Throwable, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation44, Mark4TypeLouvainCommunity4, Mark4TypeLeidenCommunity11, Mark4TypeKCoreDecomposition8, Mark4TypeMaximumKCut2, Mark4TypeHDBSCAN114]                                                           | 14                    |
| 27 | org.axonframework.conversion.Converter                                                    | Converter                                          | [Type, File, Java, ByteCode, Interface, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation6, Mark4TypeLouvainCommunity2, Mark4TypeLeidenCommunity2, Mark4TypeKCoreDecompo... | 14                    |
| 28 | org.axonframework.messaging.commandhandling.CommandBus                                    | CommandBus                                         | [Type, File, Java, ByteCode, Interface, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation2, Mark4TypeLouvainCommunity3, Mark4TypeLeidenCommunity3, Mark4TypeKCoreDecomposition10, Mark4TypeMaximumKCut99, Mark4TypeHDBSCAN-1]                                                                   | 13                    |
| 29 | org.axonframework.messaging.queryhandling.QueryMessage                                    | QueryMessage                                       | [Type, File, Java, ByteCode, Interface, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation48, Mark4TypeLouvainCommunity18, Mark4TypeLeidenCommunity8, Mark4TypeKCoreDecomposition10... | 13                    |
| 30 | org.axonframework.messaging.commandhandling.CommandResultMessage                          | CommandResultMessage                               | [Type, File, Java, ByteCode, Interface, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation2, Mark4TypeLouvainCommunity3, Mark4TypeLeidenCommunity3, Mark4TypeKCoreDecomposition10, ... | 12                    |
| 31 | org.axonframework.common.configuration.ComponentDefinition                                | ComponentDefinition                                | [Type, File, Java, ByteCode, Interface, GenericDeclaration, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation30, Mark4TypeLouvainCommunity6, Mark4TypeLeidenCommunity6, Mark4TypeKCoreDecomposition10, Mark4TypeMaximumKCut59, Mark4TypeLocalClusteringCoefficient0.2028985507246377, Mark4T... | 12                    |
| 32 | org.axonframework.common.configuration.ComponentDefinition\$IncompleteComponentDefinition | ComponentDefinition\$IncompleteComponentDefinition | [Type, File, Java, ByteCode, Interface, GenericDeclaration, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation30, Mark4TypeLouvainCommunity6, Mark4TypeLeidenCommunity6, Mark4TypeKCoreDecomposition10, Mark4TypeMaximumKCut64, Mark4TypeHDBSCAN101]                                             | 12                    |
| 33 | org.axonframework.messaging.core.Context                                                  | Context                                            | [Type, File, Java, ByteCode, Interface, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityBetweenness, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation11, Mark4TypeLouvainCommunity1, Mark4TypeLeidenC... | 12                    |
| 34 | org.axonframework.common.ReflectionUtils                                                  | ReflectionUtils                                    | [Type, File, Java, ByteCode, Class, Mark4TopCentralityBetweenness, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation2, Mark4TypeLouvainCommunity11, Mark4TypeLeidenCommunity11, Mark4TypeKCoreDecomposition8, Mark4TypeMaximumKCut20, Mark4TypeHDBSCAN114, Mark4TopAnomalyBottleneck]           | 12                    |
| 35 | org.axonframework.common.AxonNonTransientException                                        | AxonNonTransientException                          | [Type, File, Java, ByteCode, Class, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TypeWeaklyConnectedComponent0,                                                                                                                                                                          | 11                    |

|    | fullQualifiedDependentTypeName                                      | dependentTypeName         | dependentTypeLabels                                                                                                                                                                                                                                                                                                           | numberOfUsingPackages |
|----|---------------------------------------------------------------------|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
|    |                                                                     |                           | Mark4TypeLabelPropagation4,<br>Mark4TypeLouvainCommunity9,<br>Mark4TypeLeidenCommunity10,<br>Mark4TypeKCoreDecomposition4,<br>Mark4TypeMaximumKCut72,<br>Mark4TypeLocalClusteringCoe...                                                                                                                                       |                       |
| 36 | org.axonframework.common.configuration.ComponentBuilder             | ComponentBuilder          | [Type, File, Java, ByteCode, Interface, GenericDeclaration,<br>Mark4TypeWeaklyConnectedComponent0,<br>Mark4TypeLabelPropagation16,<br>Mark4TypeLouvainCommunity6,<br>Mark4TypeLeidenCommunity6,<br>Mark4TypeKCoreDecomposition10,<br>Mark4TypeMaximumKCut84, Mark4TypeHDBSCAN-1]                                              | 11                    |
| 37 | org.axonframework.messaging.eventhandling.conversion.EventConverter | EventConverter            | [Type, File, Java, ByteCode, Interface,<br>Mark4TypeWeaklyConnectedComponent0,<br>Mark4TypeLabelPropagation8,<br>Mark4TypeLouvainCommunity12,<br>Mark4TypeLeidenCommunity3,<br>Mark4TypeKCoreDecomposition9,<br>Mark4TypeMaximumKCut72, Mark4TypeHDBSCAN-1]                                                                   | 11                    |
| 38 | org.axonframework.messaging.core.MessageHandlerInterceptor          | MessageHandlerInterceptor | [Type, File, Java, ByteCode, Interface, GenericDeclaration,<br>Mark4TopCentralityBetweenness,<br>Mark4TopCentralityHyperlinkInducedTopicSearchAuthority,<br>Mark4TopCentralityHyperlinkInducedTopicSearchHub,<br>Mark4TypeWeaklyConnectedComponent0,<br>Mark4TypeLabelPropagation50,<br>Mark4TypeLouvainCommunity6, Mark4T... | 11                    |
| 39 | org.axonframework.messaging.core.annotation.ParameterResolver       | ParameterResolver         | [Type, File, Java, ByteCode, Interface, GenericDeclaration,<br>Mark4TypeWeaklyConnectedComponent0,<br>Mark4TypeLabelPropagation5,<br>Mark4TypeLouvainCommunity4,<br>Mark4TypeLeidenCommunity11,<br>Mark4TypeKCoreDecomposition7,<br>Mark4TypeMaximumKCut62, Mark4TypeHDBSCAN78]                                               | 11                    |

Table 6 - Packages that are used by multiple artifacts

This table shows the top 30 artifacts that only use a few (compared to all existing) packages of another artifact. The whole table can be found in the CSV report [ArtifactPackageUsage](#) .

|    | artifactName                         | dependentArtifactName         | dependentPackages | dependentArtifactPackages | packageUsagePercentage | dependentFullQualifiedPackageNames                                                                                                                                                                                                                                                                           | dependentPackageNamees                                                                                                                                                   |
|----|--------------------------------------|-------------------------------|-------------------|---------------------------|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0  | axon-tracing-opentelemetry-5.0.2     | axon-messaging-5.0.2          | 2                 | 57                        | 0.035088               | [org.axonframework.messaging.tracing, org.axonframework.messaging.core]                                                                                                                                                                                                                                      | [tracing, core]                                                                                                                                                          |
| 1  | axon-tracing-opentelemetry-5.0.2     | axon-common-5.0.2             | 1                 | 15                        | 0.066667               | [org.axonframework.common]                                                                                                                                                                                                                                                                                   | [common]                                                                                                                                                                 |
| 2  | axon-metrics-micrometer-5.0.2        | axon-messaging-5.0.2          | 6                 | 57                        | 0.105263               | [org.axonframework.messaging.core, org.axonframework.messaging.eventhandling, org.axonframework.messaging.queryhandling, org.axonframework.messaging.monitoring, org.axonframework.messaging.commandhandling, org.axonframework.messaging.eventhandling.processing]                                          | [core, eventhandling, queryhandling, monitoring, commandhandling, processing]                                                                                            |
| 3  | axon-test-5.0.2                      | axon-eventsourcing-5.0.2      | 1                 | 8                         | 0.125000               | [org.axonframework.eventsourcing.eventstore]                                                                                                                                                                                                                                                                 | [eventstore]                                                                                                                                                             |
| 4  | axon-server-connector-5.0.2          | axon-eventsourcing-5.0.2      | 1                 | 8                         | 0.125000               | [org.axonframework.eventsourcing.eventstore]                                                                                                                                                                                                                                                                 | [eventstore]                                                                                                                                                             |
| 5  | axon-metrics-micrometer-5.0.2        | axon-common-5.0.2             | 2                 | 15                        | 0.133333               | [org.axonframework.common.configuration, org.axonframework.common]                                                                                                                                                                                                                                           | [configuration, common]                                                                                                                                                  |
| 6  | axon-server-connector-5.0.2          | axon-modelling-5.0.2          | 1                 | 7                         | 0.142857               | [org.axonframework.modelling]                                                                                                                                                                                                                                                                                | [modelling]                                                                                                                                                              |
| 7  | axon-test-5.0.2                      | axon-messaging-5.0.2          | 9                 | 57                        | 0.157895               | [org.axonframework.messaging.commandhandling, org.axonframework.messaging.core, org.axonframework.messaging.eventhandling, org.axonframework.messaging.eventstreaming, org.axonframework.messaging.core.conversion, org.axonframework.messaging.eventhandling.processing.streaming.token, org.axonframew...  | [commandhandling, core, eventhandling, eventstreaming, conversion, token, unitofwork, monitoring, annotation]                                                            |
| 8  | axon-spring-boot-autoconfigure-5.0.2 | axon-test-5.0.2               | 1                 | 5                         | 0.200000               | [org.axonframework.test.server]                                                                                                                                                                                                                                                                              | [server]                                                                                                                                                                 |
| 9  | axon-update-5.0.2                    | axon-common-5.0.2             | 3                 | 15                        | 0.200000               | [org.axonframework.common.annotation, org.axonframework.common.configuration]                                                                                                                                                                                                                                | [annotation, common, configuration]                                                                                                                                      |
| 10 | axon-spring-boot-autoconfigure-5.0.2 | axon-server-connector-5.0.2   | 1                 | 5                         | 0.200000               | [org.axonframework.axonserver.connector]                                                                                                                                                                                                                                                                     | [connector]                                                                                                                                                              |
| 11 | axon-conversion-5.0.2                | axon-common-5.0.2             | 3                 | 15                        | 0.200000               | [org.axonframework.common, org.axonframework.common.infra, org.axonframework.common.annotation]                                                                                                                                                                                                              | [common, infra, annotation]                                                                                                                                              |
| 12 | axon-modelling-5.0.2                 | axon-messaging-5.0.2          | 13                | 57                        | 0.228070               | [org.axonframework.messaging.eventhandling, org.axonframework.messaging.conversion, org.axonframework.messaging.core.unitofwork, org.axonframework.messaging.core, org.axonframework.messaging.core.annotation, org.axonframework.messaging.commandhandling.annotation, org.axonframework....]               | [eventhandling, conversion, unitofwork, core, annotation, reflection, commandhandling, configuration]                                                                    |
| 13 | axon-modelling-5.0.2                 | axon-conversion-5.0.2         | 1                 | 4                         | 0.250000               | [org.axonframework.conversion]                                                                                                                                                                                                                                                                               | [conversion]                                                                                                                                                             |
| 14 | axon-spring-boot-autoconfigure-5.0.2 | axon-eventsourcing-5.0.2      | 2                 | 8                         | 0.250000               | [org.axonframework.eventsourcing.eventstore, org.axonframework.eventsourcing.eventstore.jpa]                                                                                                                                                                                                                 | [eventstore, jpa]                                                                                                                                                        |
| 15 | axon-eventsourcing-5.0.2             | axon-conversion-5.0.2         | 1                 | 4                         | 0.250000               | [org.axonframework.conversion]                                                                                                                                                                                                                                                                               | [conversion]                                                                                                                                                             |
| 16 | axon-conversion-5.0.2                | axon-conversion-5.0.2         | 1                 | 4                         | 0.250000               | [org.axonframework.conversion]                                                                                                                                                                                                                                                                               | [conversion]                                                                                                                                                             |
| 17 | axon-spring-boot-autoconfigure-5.0.2 | axon-messaging-5.0.2          | 15                | 57                        | 0.263158               | [org.axonframework.messaging.core.timeout, org.axonframework.messaging.core.conversion, org.axonframework.messaging.core.correlation, org.axonframework.messaging.eventhandling, org.axonframework.messaging.commandhandling, org.axonframework.messaging.queryhandling.distributed, org.axonframework.m...] | [timeout, conversion, correlation, eventhandling, commandhandling, distributed, store, interception, transaction, jpa, queryhandling, core, annotation]                  |
| 18 | axon-test-5.0.2                      | axon-common-5.0.2             | 4                 | 15                        | 0.266667               | [org.axonframework.common, org.axonframework.common.annotation, org.axonframework.common.configuration, org.axonframework.common.infra]                                                                                                                                                                      | [common, annotation, configuration, infra]                                                                                                                               |
| 19 | axon-server-connector-5.0.2          | axon-messaging-5.0.2          | 16                | 57                        | 0.280702               | [org.axonframework.messaging.eventhandling.processing.streaming, org.axonframework.messaging.eventhandling.conversion, org.axonframework.messaging.core.unitofwork, org.axonframework.messaging.core, org.axonframework.messaging.eventhandling.processing, org.axonframework.messaging.eventhandling.pr...  | [streaming, conversion, unitofwork, core, processing, segmenting, token, subscribing, eventhandling, eventstreaming, store, distributed, commandhandling, queryhandling] |
| 20 | axon-eventsourcing-5.0.2             | axon-messaging-5.0.2          | 16                | 57                        | 0.280702               | [org.axonframework.messaging.core.unitofwork, org.axonframework.messaging.core.unitofwork.transaction, org.axonframework.messaging.core, org.axonframework.messaging.eventhandling, org.axonframework.messaging.eventhandling.conver...]                                                                     | [unitofwork, transaction, core, eventhandling, token, conversion, eventstreaming, annotation, commandhandling, configuration, interception]                              |
| 21 | axon-spring-boot-autoconfigure-5.0.2 | axon-common-5.0.2             | 5                 | 15                        | 0.333333               | [org.axonframework.common.jpa, org.axonframework.common.annotation, org.axonframework.common, org.axonframework.common.jdbc, org.axonframework.common.configuration]                                                                                                                                         | [jpa, annotation, common, jdbc, configuration]                                                                                                                           |
| 22 | axon-modelling-5.0.2                 | axon-common-5.0.2             | 5                 | 15                        | 0.333333               | [org.axonframework.common, org.axonframework.common.configuration, org.axonframework.common.infra, org.axonframework.common.property, org.axonframework.common.annotation]                                                                                                                                   | [common, configuration, infra, property, annotation]                                                                                                                     |
| 23 | axon-server-connector-5.0.2          | axon-common-5.0.2             | 6                 | 15                        | 0.400000               | [org.axonframework.common, org.axonframework.common.util, org.axonframework.common.annotation, org.axonframework.common.configuration, org.axonframework.common.infra, org.axonframework.common.lifecycle]                                                                                                   | [common, util, annotation, configuration, infra, lifecycle]                                                                                                              |
| 24 | axon-metrics-micrometer-5.0.2        | axon-metrics-micrometer-5.0.2 | 1                 | 2                         | 0.500000               | [org.axonframework.extension.metrics.micrometer.reservoir]                                                                                                                                                                                                                                                   | [reservoir]                                                                                                                                                              |
| 25 | axon-messaging-5.0.2                 | axon-conversion-5.0.2         | 2                 | 4                         | 0.500000               | [org.axonframework.conversion, org.axonframework.conversion.json]                                                                                                                                                                                                                                            | [conversion, json]                                                                                                                                                       |
| 26 | axon-common-5.0.2                    | axon-common-5.0.2             | 8                 | 15                        | 0.533333               | [org.axonframework.common.annotation, org.axonframework.common.function, org.axonframework.common, org.axonframework.common.lifecycle, org.axonframework.common.infra, org.axonframework.common.tx, org.axonframework.common.io, org.axonframework.common.configuration]                                     | [annotation, function, common, lifecycle, infra, tx, io, configuration]                                                                                                  |
| 27 | axon-eventsourcing-5.0.2             | axon-common-5.0.2             | 9                 | 15                        | 0.600000               | [org.axonframework.common.tx, org.axonframework.common.annotation, org.axonframework.common.jdbc, org.axonframework.common.infra, org.axonframework.common.function, org.axonframework.common.jpa, org.axonframework.common, org.axonframework.common.io, org.axonframework.common.configuration]            | [tx, annotation, jdbc, infra, function, jpa, common, io, configuration]                                                                                                  |
| 28 | axon-test-5.0.2                      | axon-test-5.0.2               | 3                 | 5                         | 0.600000               | [org.axonframework.test, org.axonframework.test.matchers, org.axonframework.test.util]                                                                                                                                                                                                                       | [test, matchers, util]                                                                                                                                                   |
| 29 | axon-messaging-5.0.2                 | axon-common-5.0.2             | 9                 | 15                        | 0.600000               | [org.axonframework.common, org.axonframework.common.annotation, org.axonframework.common.infra, org.axonframework.common.util, org.axonframework.common.jdbc, org.axonframework.common.property, org.axonframework.common.tx, org.axonframework.common.configuration, org.axonframework.common.jpa]          | [common, annotation, infra, util, jdbc, property, tx, configuration, jpa]                                                                                                |

Table 7 - Types that are used by multiple artifacts

This table shows the top 30 types that only use a few (compared to all existing) types of another artifact. The whole table can be found in the CSV report `ClassesPerPackageUsageAcrossArtifacts`.

|    | artifactName                         | dependentArtifactName | packageName                                                                | dependentPackage.fqn                        | dependentTypes | dependentPackageTypes | typeUsagePercentag |
|----|--------------------------------------|-----------------------|----------------------------------------------------------------------------|---------------------------------------------|----------------|-----------------------|--------------------|
| 0  | axon-eventsourcing-5.0.2             | axon-messaging-5.0.2  | org.axonframework.eventsourcing.configuration                              | org.axonframework.messaging.core            | 1              | 80                    | 0.01250            |
| 1  | axon-test-5.0.2                      | axon-messaging-5.0.2  | org.axonframework.test.matchers                                            | org.axonframework.messaging.core            | 1              | 80                    | 0.01250            |
| 2  | axon-modelling-5.0.2                 | axon-messaging-5.0.2  | org.axonframework.modelling.repository                                     | org.axonframework.messaging.core            | 1              | 80                    | 0.01250            |
| 3  | axon-eventsourcing-5.0.2             | axon-messaging-5.0.2  | org.axonframework.eventsourcing.eventstore.jdbc                            | org.axonframework.messaging.core            | 1              | 80                    | 0.01250            |
| 4  | axon-spring-boot-autoconfigure-5.0.2 | axon-messaging-5.0.2  | org.axonframework.extension.springboot.autoconfig                          | org.axonframework.messaging.core.annotation | 1              | 50                    | 0.02000            |
| 5  | axon-eventsourcing-5.0.2             | axon-messaging-5.0.2  | org.axonframework.eventsourcing.configuration                              | org.axonframework.messaging.core.annotation | 1              | 50                    | 0.02000            |
| 6  | axon-messaging-5.0.2                 | axon-common-5.0.2     | org.axonframework.messaging.core                                           | org.axonframework.common.configuration      | 1              | 46                    | 0.02173            |
| 7  | axon-eventsourcing-5.0.2             | axon-common-5.0.2     | org.axonframework.eventsourcing.annotation.reflection                      | org.axonframework.common.configuration      | 1              | 46                    | 0.02173            |
| 8  | axon-modelling-5.0.2                 | axon-common-5.0.2     | org.axonframework.modelling.entity.annotation                              | org.axonframework.common.configuration      | 1              | 46                    | 0.02173            |
| 9  | axon-eventsourcing-5.0.2             | axon-common-5.0.2     | org.axonframework.eventsourcing.annotation                                 | org.axonframework.common.configuration      | 1              | 46                    | 0.02173            |
| 10 | axon-messaging-5.0.2                 | axon-common-5.0.2     | org.axonframework.messaging.core.unitofwork                                | org.axonframework.common.configuration      | 1              | 46                    | 0.02173            |
| 11 | axon-metrics-micrometer-5.0.2        | axon-messaging-5.0.2  | org.axonframework.extension.metrics.micrometer                             | org.axonframework.messaging.core            | 2              | 80                    | 0.02500            |
| 12 | axon-tracing-opentelemetry-5.0.2     | axon-messaging-5.0.2  | org.axonframework.extension.tracing.opentelemetry                          | org.axonframework.messaging.core            | 2              | 80                    | 0.02500            |
| 13 | axon-conversion-5.0.2                | axon-common-5.0.2     | org.axonframework.conversion.avro                                          | org.axonframework.common                    | 1              | 34                    | 0.02941            |
| 14 | axon-messaging-5.0.2                 | axon-common-5.0.2     | org.axonframework.messaging.eventhandling.processing.streaming.token.store | org.axonframework.common                    | 1              | 34                    | 0.02941            |
| 15 | axon-messaging-5.0.2                 | axon-common-5.0.2     | org.axonframework.messaging.commandhandling.tracing                        | org.axonframework.common                    | 1              | 34                    | 0.02941            |
| 16 | axon-messaging-5.0.2                 | axon-common-5.0.2     | org.axonframework.messaging.core.timeout                                   | org.axonframework.common                    | 1              | 34                    | 0.02941            |
| 17 | axon-tracing-opentelemetry-5.0.2     | axon-common-5.0.2     | org.axonframework.extension.tracing.opentelemetry                          | org.axonframework.common                    | 1              | 34                    | 0.02941            |
| 18 | axon-messaging-5.0.2                 | axon-common-5.0.2     | org.axonframework.messaging.queryhandling.tracing                          | org.axonframework.common                    | 1              | 34                    | 0.02941            |
| 19 | axon-messaging-5.0.2                 | axon-common-5.0.2     | org.axonframework.messaging.eventhandling.processing.streaming.segmenting  | org.axonframework.common                    | 1              | 34                    | 0.02941            |
| 20 | axon-eventsourcing-5.0.2             | axon-common-5.0.2     | org.axonframework.eventsourcing.annotation                                 | org.axonframework.common                    | 1              | 34                    | 0.02941            |
| 21 | axon-messaging-5.0.2                 | axon-common-5.0.2     | org.axonframework.messaging.core.interception.annotation                   | org.axonframework.common                    | 1              | 34                    | 0.02941            |
| 22 | axon-messaging-5.0.2                 | axon-common-5.0.2     | org.axonframework.messaging.eventhandling.processing                       | org.axonframework.common                    | 1              | 34                    | 0.02941            |
| 23 | axon-spring-boot-autoconfigure-5.0.2 | axon-common-5.0.2     | org.axonframework.extension.springboot.autoconfig                          | org.axonframework.common                    | 1              | 34                    | 0.02941            |
| 24 | axon-test-5.0.2                      | axon-common-5.0.2     | org.axonframework.test.util                                                | org.axonframework.common                    | 1              | 34                    | 0.02941            |
| 25 | axon-messaging-5.0.2                 | axon-common-5.0.2     | org.axonframework.messaging.monitoring.configuration                       | org.axonframework.common                    | 1              | 34                    | 0.02941            |
| 26 | axon-messaging-5.0.2                 | axon-common-5.0.2     | org.axonframework.messaging.eventhandling.sequencing                       | org.axonframework.common                    | 1              | 34                    | 0.02941            |
| 27 | axon-messaging-5.0.2                 | axon-common-5.0.2     | org.axonframework.messaging.monitoring                                     | org.axonframework.common                    | 1              | 34                    | 0.02941            |
| 28 | axon-messaging-5.0.2                 | axon-common-5.0.2     | org.axonframework.messaging.eventhandling.tracing                          | org.axonframework.common                    | 1              | 34                    | 0.02941            |
| 29 | axon-test-5.0.2                      | axon-common-5.0.2     | org.axonframework.test.server                                              | org.axonframework.common                    | 1              | 34                    | 0.02941            |

Table 8 - Duplicate package names across artifacts

This table shows the top 30 duplicate package names across artifacts. They are ordered by the number of duplicates descending.

This might lead to confusion, makes importing more error prone and might even lead to duplicate classes where only one of them will be loaded by the class loader. If a package is named the same way in two or more artifacts this even allows another artifact to access package protected classes, methods or members which might not be intended.

The whole table can be found in the CSV report `DuplicatePackageNamesAcrossArtifacts`.

| packageName | duplicates | artifactNames |
|-------------|------------|---------------|
|-------------|------------|---------------|

Table 9 - Annotated elements

This table shows 30 most used Java Annotations including some examples where they are used.

|    | annotationName                                                            | languageElement | numberOfAnnotatedElements |                                                                                                                                                                                                                                 |
|----|---------------------------------------------------------------------------|-----------------|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0  | jakarta.annotation.NonNull                                                | Parameter       | 3440                      | [org.axonframework.test.fixture.RecordingCommandBus.<init>(0), org.axonframework.test.fixture.Re<br>org.axonframework.test.fixture.RecordingCommandBus.subscribe(0), org.axonframework.test.fixture.Rec<br>org.axon             |
| 1  | jakarta.annotation.NonNull                                                | Method          | 837                       | [org.axonframework.test.FixtureResourceParameterResolverFactory\$FailingParamet<br>org.axonframework.update.UpdateChecker.buildRequest, org.axonframework.update.api<br>org.axonframework.update.api.UpdateCheck                |
| 2  | jakarta.annotation.Nullable                                               | Parameter       | 358                       | [org.axonframework.test.fixture.RecordingCommandBus.dispatch(1), org.axonframework.test.fixture.<br>org.axonframework.test.fixture.AxonTestThenNothing.<init>(4), org.axonframework.test.fixture.<br>org.axonframew             |
| 3  | jakarta.annotation.Nullable                                               | Method          | 99                        | [org.axonframework.test.fixture<br>org.axonframework.messaging.commandhandling.annotation.CommandDispatcherParamet<br>org.axonfram                                                                                              |
| 4  | org.axonframework.common.annotation.Internal                              | Class           | 79                        | [org.axonframework.test.fixture.RecordingCommandBus, org.axonframework<br>org.axonframework.test.util.RecordingCommandBus, org.axonframework<br>org.axonframework.test.fixture.RecordingEventStore, org.axonfram                |
| 5  | jakarta.annotation.NonNull                                                | Field           | 72                        | [org.axonframework.test.fixture.AxonTestFixturesCustomization.fieldFilters, org.axonframework.update<br>org.axonframework.update.api.DetectedVulnerability.artifactId, org.axonframework.update<br>org.axon                     |
| 6  | java.lang.FunctionalInterface                                             | Interface       | 55                        | [org.axonframework.test.matchers.FieldFilter, org.axonframework.update.configuration.Environm<br>EnvironmentVariableSupplier, org.axonframework.messaging.commandhandling.distribut<br>Handler, org.axonframework.messaging.cor |
| 7  | java.lang.annotation.Target                                               | Annotation      | 42                        | [org.axonframework.extension.springboot.util.RegisterDefaultEntities, org.axonframework.extension.springboot.<br>org.axonframework.extension.springboot.util.Ci<br>org.axonframework.messaging.commandhandling.an               |
| 8  | java.lang.annotation.Retention                                            | Annotation      | 42                        | [org.axonframework.extension.springboot.util.RegisterDefaultEntities, org.axonframework.extension.springboot.<br>org.axonframework.extension.springboot.util.Ci<br>org.axonframework.messaging.commandhandling.an               |
| 9  | org.springframework.context.annotation.Bean                               | Method          | 38                        | [org.axonframework.extension.springboot.autoconfigure.UpdateCheckerAutoConfiguration.spr<br>org.axonframework.extension.springboot.autoconfigure.JdbcTransactionAutoConf<br>org.axonframework.extension.springboot.auto         |
| 10 | org.springframework.boot.autoconfigure.condition.ConditionalOnMissingBean | Method          | 20                        | [org.axonframework.extension.springboot.autoconfigure.JdbcTransactionAutoConf<br>org.axonframework.extension.springboot.autoconfigure.JpaAutoCor<br>org.axonframework.extension.springboot.autoconfigure.JpaAutoConf            |
| 11 | org.axonframework.common.annotation.Internal                              | Constructor     | 19                        | [org.axonframework.messaging.eventhandling.processing.streaming.pooled.PooledStreamingEv<br>org.axonframework.messaging.eventhandling.processing.streaming.pooled.PooledStreaming<br>org.axonframework.messag                   |
| 12 | org.springframework.boot.autoconfigure.AutoConfiguration                  | Class           | 19                        | [org.axonframework.extension.springboot.autoconfigure.<br>org.axonframework.extension.springboot.autoconfigure.<br>org.axonframework.extension.springboot.autoconfigure.AxonTimeoutAutoConfiguration, org.axonframework.        |
| 13 | org.axonframework.common.annotation.Internal                              | Interface       | 16                        | [org.axonframework.update.cor<br>org.axonframework.messaging.commandhandling.anno<br>org.axonframework.messaging.eventhandling.i<br>org.axonframework.messaging.queryhandling.annotation                                        |
| 14 | java.lang.annotation.Documented                                           | Annotation      | 15                        | [org.axonframework.extension.springboot.<br>org.axonframework.extension.springboot.util.Ci<br>org.axonframework.messaging.commandhand<br>org.axonframework.messaging.eventhandling.                                             |
| 15 | com.fasterxml.jackson.annotation.JsonProperty                             | Parameter       | 12                        | [org.axonframework.messaging.eventhandling.processing.streaming.token.i<br>org.axonframework.messaging.eventhandling.processing.streaming.token.i<br>org.axonframework.messaging.eventhandling.processing.stream                |
| 16 | org.axonframework.common.Priority                                         | Class           | 12                        | [org.axonframework.test.fixture.R<br>org.axonframework.messaging.eventhandling.annotation.Sequence<br>org.axonframework.messaging.eventhandling.annotation.Tim<br>org.axonframework.r                                           |
| 17 | org.axonframework.common.annotation.Internal                              | Method          | 11                        | [org.axonframework.extension.springboot.Event<br>org.axonframework.messaging.eventhandling.conversion.D<br>org.axonframework.messaging.eventhandling.processing.streaming.token.store.jdbc.JdbcTokenStore.con                   |
| 18 | org.springframework.boot.context.properties.ConfigurationProperties       | Class           | 10                        | [org.axonframework.extension.springboot.TokenStoreProperties, org.axonframework.extension.spring<br>org.axonframework.extension.springboot.DistributedCommandBusProperties, org.axonframework.exten<br>org.axon                 |
| 19 | org.springframework.boot.context.properties.EnableConfigurationProperties | Class           | 9                         | [org.axonframework.extension.springboot.autoconfigure.<br>org.axonframework.extension.springboot.autoconfigure.<br>org.axonframework.extension.springboot.autoconfigure.ConverterAutoConfiguration, org.axonframework.exten     |
| 20 | org.springframework.boot.autoconfigure.condition.ConditionalOnProperty    | Class           | 8                         | [org.axonframework.extension.springboot.autoconfigure.<br>org.axonframework.extension.springboot.autoconfigure.AvroSchemaStoreAutoConfiguration.AvroConfiguredC<br>org.axonframework.extension.springboot.autoconfigure.AvroSc  |
| 21 | org.springframework.boot.context.properties.NestedConfigurationProperty   | Field           | 8                         | [org.axonframework.exten<br>MessageHandlerTimeoutProperties.events, org.axonframework.extension.spr<br>MessageHan<br>org.axonframework.extension.springboot.TimeoutProperties\$MessageHan                                       |
| 22 | jakarta.persistence.Basic                                                 | Field           | 8                         | [org.axonframework.messaging.eventhandling.processing.streaming.token<br>org.axonframework.messaging.eventhandling.processing.streaming.toke<br>org.axonframework.messaging.eventhandling.processing.streaming                  |
| 23 | java.lang.SafeVarargs                                                     | Constructor     | 7                         | [org.axonframework.test.matchers.ExactSequenceMatcher.<init>, org.axonframework.test.mat<br>org.axonframework.test.matchers.SequenceMatcher.<init>, org.axonframework.test.matcl<br>org.axonframework.test.r                    |
| 24 | org.springframework.boot.autoconfigure.condition.ConditionalOnClass       | Class           | 7                         | [org.axonframework.extension.springboot.autoconfigure.A<br>org.axonframework.extension.springboot.<br>org.axonframework.extension.springboot.autocor<br>org.axonframework.extension.sp                                          |
| 25 | java.lang.Deprecated                                                      | Method          | 7                         | [org.axonframework.test.fixture.AxonTestThenCommand.resultMessagePayloadSatisfies, org.axonframework<br>Command.resultMessagePayloadSatisfies, org.axonframework.messaging.core.Generi<br>org.axonframework.messa               |
| 26 | java.beans.ConstructorProperties                                          | Constructor     | 6                         | [org.axonframework.messaging.eventhandling.processing.streaming.token<br>org.axonframework.messaging.eventhandling.processing.streaming<br>org.axonframework.messaging.eventhandling.processing.streaming.token.R               |
| 27 | org.springframework.boot.autoconfigure.AutoConfigureBefore                | Class           | 6                         | [org.axonframework.extension.springboot.autoc<br>org.axonframework.extension.springboot.autoconfigure.Correlati<br>org.axonframework.extension.springboot.autoconfigure.AvroSchemaStoreAutoConfiguration, org.axonfra           |
| 28 | org.springframework.context.annotation.Conditional                        | Method          | 6                         | [org.axonframework.extension.springboot.autoconfigure.UpdateCheckerAutoConfiguration.spr<br>org.axonframework.extension.springboot.autoconfigure.AvroSchemaStoreAutoConf<br>org.axonframework.extension.springboot.autoc        |
| 29 | org.springframework.boot.context.properties.bind.DefaultValue             | Parameter       | 6                         | [org.axonframework.extension.springboot.JpaEventStorageEngine<br>org.axonframework.extension.springboot.JpaEventStorageEngine<br>org.axonframework.extension.springboot.JpaEventStorageEngineConfiguratio                       |

Table 10 - Distance distribution between dependent files

This table shows the file directory distance distribution between dependent files. Intuitively, the distance is given by the fewest number of change directory commands needed to navigate between a file and a dependency it uses. Those are aggregate to see how many dependent files are in the same directory, how many are just one change directory command apart, and so on.



| dependency.fileDistanceAsFewestChangeDirectoryCommands | numberOfDependencies | numberOfDependencyUsers | numberOfDependencyProviders |     |                                                                                                                                            |
|--------------------------------------------------------|----------------------|-------------------------|-----------------------------|-----|--------------------------------------------------------------------------------------------------------------------------------------------|
| 0                                                      | 0                    | 2090                    | 863                         | 904 | [/axon-spring-boot-autoconfigure-5.0.2.jar uses /axon-test<br>/org/axonframework/test/fixture uses /org/axonframew                         |
| 1                                                      | 1                    | 96                      | 82                          | 41  | [/org/axonframework/test/fixture uses /org/axonframework/test, /org/axonframew<br>uses /org/axonframework/update/dete                      |
| 2                                                      | 2                    | 2059                    | 609                         | 411 | Customization.classuses/org/axonframework/test/matchers/Fi<br>/CommandValidator.classuses/org/axonframework/test/match<br>/AxonTestFixture |
| 3                                                      | 4                    | 2073                    | 666                         | 317 | [/org/axonframework/e<br>/org/axonframework/extension/springboot/service/connection/AxonServerTestCon                                      |